

Task 1

The Booking_receipt table violates 1NF because the seat_numbers attribute can contain multiple values in one field, for example "12A, 12B".

In 1NF, each attribute should contain only one atomic value. Therefore, each seat number should be stored separately.

The table may also violate 2NF and 3NF because some attributes may depend on only part of a key or on other non-key attributes. These dependencies will be identified during the normalization process in the following tasks.

Task 2 — 1NF

To convert Booking_receipt into 1NF, the multi-valued seat_numbers attribute is replaced by a single-valued seat_number attribute.

For example:

"12A, 12B" → separate rows containing 12A and 12B.

The resulting table is:

Booking_receipt_1NF

- booking_id
- passenger_full_name
- passenger_passport_number
- flight_number
- departure_airport_name
- departure_city
- arrival_airport_name
- arrival_city
- airline_name
- seat_number
- ticket_price

Assumption for the normalization process:

Primary Key: (booking_id, seat_number)

At this stage, there are no foreign keys because the data is still kept in one table.

Task 2 — 2NF

The 1NF table uses the composite primary key:

(booking_id, seat_number)

However, some attributes depend only on booking_id rather than on the whole composite key.

For example:

booking_id → flight_number

This is a partial dependency, so the table is not in 2NF.

To remove the partial dependency, the table is divided into:

Booking_2NF

- booking_id **PK**
- flight_number
- departure_airport_name
- departure_city
- arrival_airport_name
- arrival_city
- airline_name
- ticket_price

Booking_Seat_2NF

- booking_id **PK, FK**
- seat_number **PK**
- passenger_full_name
- passenger_passport_number

Foreign Key:

Booking_Seat_2NF.booking_id → Booking_2NF.booking_id

Task 2 — 3NF

The 2NF tables may still contain transitive dependencies.

For example:

booking_id → flight_number

flight_number → airline_name

Therefore, airline_name depends indirectly on booking_id through flight_number.

To reduce this transitive dependency, the data can be further separated into the

following tables.

Booking

- booking_id **PK**
- flight_number **FK**
- ticket_price

Flight

- flight_number **PK**
- airline_name
- departure_airport_name
- arrival_airport_name

Airport

- airport_name **PK**
- city

Passenger

- passenger_passport_number **PK**
- passenger_full_name

Booking_Seat

- booking_id **PK, FK**
- seat_number **PK**
- passenger_passport_number **FK**

Foreign Keys:

Booking.flight_number → Flight.flight_number

Booking_Seat.booking_id → Booking.booking_id

**Booking_Seat.passenger_passport_number →
Passenger.passenger_passport_number**

Assumption:

airport_name → city, so airport information is stored separately to avoid repeated city information.

Task 3 — Functional Dependencies

The following functional dependencies were used during the normalization process:

- booking_id → flight_number
- booking_id → ticket_price
- flight_number → airline_name
- passenger_passport_number → passenger_full_name
- departure_airport_name → departure_city
- arrival_airport_name → arrival_city

The first two example dependency patterns are based on the booking and flight relationships used in the normalization process.

Additional assumptions used:

- A passport number identifies one passenger.
- An airport name determines its city.
- **Task 4 — Flights Table and 3NF**
- The primary key of the `Flights` table is:
- **flight_id**
- The main functional dependency is:
- flight_id → sch_departure_time, sch_arrival_time, departing_airport_id, arriving_airport_id, departing_gate, arriving_gate, airline_id, act_departure_time, act_arrival_time, created_at, updated_at
- The `Flights` table stores only the IDs of related airports and airlines.
- For example:
- departing_airport_id refers to the `Airport` table, while airline_id refers to the `Airline` table.
- Airport details such as airport_name and city, and airline details such as airline_name and airline_country, are not stored directly in the `Flights` table.
- Therefore, there is no obvious transitive dependency between non-key attributes inside the `Flights` table.
- **Conclusion: the Flights table is in 3NF**

DDL Task 1: Create Tables

The screenshot shows the pgAdmin 4 interface. The query editor contains the following SQL:

```
SELECT tablename
FROM pg_tables
WHERE schemaname = 'public';
```

The Data Output pane shows the results of the query:

tablename
passengers
security_ch...
booking
baggage_ch...
baggage
boarding_pa...
booking_flag...
flights
airport
airline

Total rows: 10 Query complete 00:00:00.079

DDL Task 4 + Task 6

The screenshot shows the pgAdmin 4 interface. The query editor contains the following SQL:

```
tc.table_name,
kcu.column_name,
ccu.table_name AS foreign_table_name,
ccu.column_name AS foreign_column_name
FROM information_schema.table_constraints AS tc
JOIN information_schema.key_column_usage AS kcu
ON tc.constraint_name = kcu.constraint_name
JOIN information_schema.constraint_column_usage AS ccu
ON ccu.constraint_name = tc.constraint_name
WHERE tc.constraint_type = 'FOREIGN KEY';

SELECT column_name, data_type, is_nullable
FROM information_schema.columns
WHERE table_name = 'booking'
ORDER BY ordinal_position;
```

The Data Output pane shows the results of the second query:

column_name	data_type	is_nullable
booking_id	integer	NO
flight_id	integer	NO
passenger_id	integer	NO
booking_platfor...	character varying	NO
created_at	timestamp without time zo...	NO
updated_at	timestamp without time zo...	NO
status	character varying	NO
ticket_price	numeric	NO

Total rows: 8 Query complete 00:00:00.050

Successfully run. Total query runtime: 50 msec. 8 rows affected.

DDL Task 7 + Task 4

pgAdmin 4

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lab2_ddl/postgres@PostgreSQL 17

Query Query History

```

283      tc.table_name,
284      kcu.column_name,
285      ccu.table_name AS foreign_table_name,
286      ccu.column_name AS foreign_column_name
287 FROM information_schema.table_constraints AS tc
288 JOIN information_schema.key_column_usage AS kcu
289   ON tc.constraint_name = kcu.constraint_name
290 JOIN information_schema.constraint_column_usage AS ccu
291   ON ccu.constraint_name = tc.constraint_name
292 WHERE tc.constraint_type = 'FOREIGN KEY';
293
294
295 SELECT column_name, data_type, is_nullable
296 FROM information_schema.columns
297 WHERE table_name = 'bookings'
298 ORDER BY ordinal_position;
299
300
301 SELECT column_name, data_type, is_nullable
302 FROM information_schema.columns
303 WHERE table_name = 'flights'
304    AND column_name = 'departing_gate';
305
306
307

```

Data Output Messages Notifications

column_name	data_type	is_nullable
name	character varying	character varying (3)
1 departing_gate	text	NO

Showing rows: 1 to 1 Page No: 1 of 1

Total rows: 1 Query complete 00:00:00.047

Successfully run. Total query runtime: 47 msec. 1 rows affected.

DDL Task 5 + Task 8

pgAdmin 4

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Query Query History

```

296 FROM information_schema.columns
297 WHERE table_name = 'bookings'
298 ORDER BY ordinal_position;
299
300
301 SELECT column_name, data_type, is_nullable
302 FROM information_schema.columns
303 WHERE table_name = 'flights'
304    AND column_name = 'departing_gate';
305
306
307 SELECT column_name
308 FROM information_schema.columns
309 WHERE table_name = 'airline'
310 ORDER BY ordinal_position;
311
312
313
314
315
316
317
318
319
320

```

Data Output Messages Notifications

column_name
name
1 airline_id
2 airline_code
3 airline_name
4 airline_country
5 created_at
6 updated_at

Showing rows: 1 to 6 Page No: 1 of 1

Total rows: 6 Query complete 00:00:00.086

Successfully run. Total query runtime: 86 msec. 6 rows affected.

DDL Task 2: Primary Keys

pgAdmin 4

File Object Tools Edit View Window Help

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lab2_ddl/postgres@PostgreSQL 17

No limit

Query Query History

```

296 FROM information_schema.columns
297 WHERE table_name = 'booking'
298 ORDER BY ordinal_position;
299
300
301 SELECT column_name, data_type, is_nullable
302 FROM information_schema.columns
303 WHERE table_name = 'flights'
304 AND column_name = 'departing_gate';
305
306 SELECT column_name
307 FROM information_schema.columns
308 WHERE table_name = 'airline'
309 ORDER BY ordinal_position;
310
311 SELECT
312     tc.table_name,
313     kcu.column_name
314 FROM information_schema.table_constraints tc
315 JOIN information_schema.key_column_usage kcu
316 ON tc.constraint_name = kcu.constraint_name
317 AND tc.table_schema = kcu.table_schema
318 WHERE tc.constraint_type = 'PRIMARY KEY'
319 AND tc.table_schema = 'public'
320 ORDER BY tc.table_name;

```

Data Output Messages Notifications

Showing rows: 1 to 10 Page No: 1 of 1

table_name	column_name
airline	airline_id
airport	airport_id
baggage	baggage_id
baggage_ch...	baggage_check...
boarding_pa...	boarding_pass...
booking	booking_id
booking_flight	booking_flightId
flights	flightId
passengers	passenger_id
security_che...	security_check...

Total rows: 10 Query complete 00:00:00.046

Successfully run. Total query runtime: 46 msec. 10 rows affected.

DDL Task 9: Foreign Key Relationships

pgAdmin 4

File Object Tools Edit View Window Help

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lab2_ddl/postgres@PostgreSQL 17

No limit

Query Query History

```

312 tc.table_name,
313 kcu.column_name
314 FROM information_schema.table_constraints tc
315 JOIN information_schema.key_column_usage kcu
316 ON tc.constraint_name = kcu.constraint_name
317 AND tc.table_schema = kcu.table_schema
318 WHERE tc.constraint_type = 'PRIMARY KEY'
319 AND tc.table_schema = 'public'
320 ORDER BY tc.table_name;
321
322 SELECT
323     tc.table_name,
324     kcu.column_name,
325     ccu.table_name AS referenced_table,
326     ccu.column_name AS referenced_column
327 FROM information_schema.table_constraints tc
328 JOIN information_schema.key_column_usage kcu
329 ON tc.constraint_name = kcu.constraint_name
330 AND tc.table_schema = kcu.table_schema
331 JOIN information_schema.constraint_column_usage ccu
332 ON ccu.constraint_name = tc.constraint_name
333 AND ccu.table_schema = tc.table_schema
334 WHERE tc.constraint_type = 'FOREIGN KEY'
335 AND tc.table_schema = 'public'
336 ORDER BY tc.table_name, kcu.column_name;

```

Data Output Messages Notifications

Showing rows: 1 to 11 Page No: 1 of 1

table_name	column_name	referenced_table	referenced_column
baggage	booking_id	booking	booking_id
baggage_ch...	booking_id	booking	booking_id
baggage_ch...	passenger_id	passengers	passenger_id
boarding_pa...	booking_id	booking	booking_id
booking	passenger_id	passengers	passenger_id
booking_flight	booking_id	booking	booking_id
booking_flight	flightId	flights	flightId
flights	airline_id	airline	airline_id
flights	arriving_airport_id	airport	airport_id
flights	departing_airport...	airport	airport_id

Total rows: 11 Query complete 00:00:00.052